

Sampling Distributions

STAT 4530/6530
Statistical Inference for Data Scientists

Statistical Inference

So far, we have reviewed basic concepts from probability. We are ultimately interested in **statistical inference**, which is the process of using probabilistic modeling of a sample of data from a larger “population” to learn about the population.

- We would like to make estimates and/or predictions pertaining to the population
- To summarize a desired aspect of the population, we would like to use an appropriate **statistic**, which is a function of the sampled data
- Examples of statistics: sample mean, sample median

Statistical Inference

We assume that each data point in our sample arises from an appropriate probability distribution. Before observing the specific value of a data point, we think of it as a random variable that can take on a range of possible values as characterized the distribution.

Similarly, before the sample of data is observed, a statistic is a random variable and thus has a distribution. In this context, the term **sampling distribution** refers to the probability distribution of a statistic.

To move towards our goal of statistical inference, we want to build on our knowledge of probability to develop useful facts about sampling distributions.

Example: Predicting Election Result from an Exit Poll

Consider an election featuring Candidate A and Candidate B, and suppose 65.5% of respondents in an exit poll reported voting for Candidate A.

- About 17 million people voted in the election, but the exit poll sampled only 2271 voters
- How can an exit poll of only about 0.01% of voters possibly give enough information to predict the winner?

Example: Predicting Election Result from an Exit Poll

To assess whether we can predict the winner from the poll, we need to decide whether 65.5% voting for Candidate A is sufficiently high that we can predict they will win.

- Suppose that only half the voters voted for Candidate A and the other half voted for Candidate B
- In this case, would it be very unusual that 65% of an exit poll sample voted for Candidate A?
- If so, we can infer that Candidate A received *more* than half of the votes (in other words, that Candidate A won)
- We will perform a simulation to try to answer this question

Example: Predicting Election Result from an Exit Poll

For simplicity, we assume simple random sampling (the sampling method for an exit poll is generally much more complex).

To simulate the votes of 2271 voters randomly chosen from the population with a 0.50 probability of voting for Candidate A, we draw one sample from a binomial distribution with $n = 2271$ “trials” and “success” probability $p = 0.50$.

We use the R command `rbinom(1, 2271, 0.50)` and get 1135.

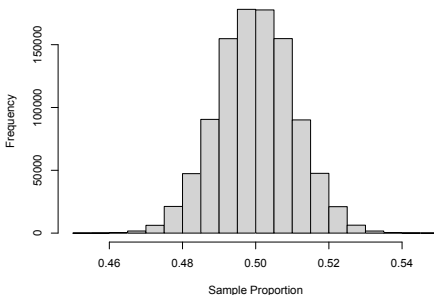
- The simulated proportion of votes for Candidate A is thus $1135/2271 = 0.4997798$.

Example: Predicting Election Result from an Exit Poll

We can perform this process one million times in order to investigate the variability of the simulated proportion in the results:

```
> results <- rbinom(1000000, 2271, 0.50)/2271  
> mean(results)  
[1] 0.4999955  
> sd(results)  
[1] 0.0104965
```

A histogram of the simulated sample proportion values shows almost all of them are between 0.46 and 0.54 (within 0.04 of the presumed portion of 0.50).



Example: Predicting Election Result from an Exit Poll

A histogram of one million simulated sample proportion values shows almost all of them are between 0.46 and 0.54 (within 0.04 of the presumed portion of 0.50).

In other words, if half of the voters had voted for Candidate A, we would have expected somewhere between 46% and 54% of voters in an exit poll of size 2271 to have voted for Candidate A.

It would therefore be extremely unusual to observe 65.5% of voters in the exit poll choosing Candidate A if exactly half of the voters voted for Candidate A. It would have been even more unusual to observe this if less than half of the voters voted for Candidate A. We therefore feel confident in predicting, based on the exit poll, that Candidate A received more than half of the votes.

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Notice that we are not being very precise in our development (“extremely unusual,” “even more unusual,” etc.). To be more precise, we will study sampling distributions in detail.

Sampling Distribution of Sample Means

Consider a sample of random variables $X_1, \dots, X_n$. The **sample mean** is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

We can compute the mean and variance of $\bar{X}$ by using the facts that

$$\begin{aligned} E\left(\sum_{i=1}^n X_i\right) &= \sum_{i=1}^n E(X_i), \\ \text{Var}\left(\sum_{i=1}^n X_i\right) &= \sum_{i=1}^n \sum_{j=1}^n \text{Cov}(X_i, X_j), \end{aligned}$$

and, if the X_i are independent,

$$\text{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \text{Cov}(X_i, X_i) = \sum_{i=1}^n \text{Var}(X_i).$$

Sampling Distribution of Sample Means

If $X_1, \dots, X_n$ are independent and arise from the same distribution, with mean μ and variance σ^2 , then

- $E(\bar{X}) = \mu$
- $Var(\bar{X}) = \frac{\sigma^2}{n}$.

The spread of a sampling distribution is characterized by its standard deviation, which is called the **standard error**. We denote the standard error of $\bar{X}$ by $\sigma_{\bar{X}}$ and note that

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}.$$

As the sample size n increases, the standard error decreases, which means that $\bar{X}$ tends to be closer to μ . So in some sense $\bar{X}$ converges to μ . This can be formalized in precise mathematical terms as the law of large numbers.

Law of Large Numbers

The **law of large numbers** states that if we have n independent observations (such as in a randomized study) from a distribution with mean μ , then for any $\epsilon > 0$,

$$P(|\bar{X} - \mu| \geq \epsilon) \rightarrow 0 \text{ as } n \rightarrow \infty$$

or, in other words, $\bar{X}$ **converges in probability** to μ .

Central Limit Theorem

We know that the sampling distribution of $\bar{X}$ narrows as the sample size n increases. But what is its shape?

The **central limit theorem** states that for a large number n of independent observations from a population distribution having mean μ and finite standard deviation σ , the sampling distribution of the sample mean $\bar{X}$ is approximately a normal distribution with mean μ and standard error $\sigma/\sqrt{n}$.

Central Limit Theorem

The **central limit theorem** states that for a large number n of independent observations from a population distribution having mean μ and finite standard deviation σ , the sampling distribution of the sample mean $\bar{X}$ is approximately a normal distribution with mean μ and standard error $\sigma/\sqrt{n}$. More precisely, the standardized version of the sample mean,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} = \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$$

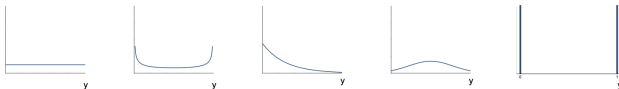
has a cdf that converges to the cdf of a standard normal distribution as $n \rightarrow \infty$. In other words, $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$ **converges in distribution** to a $N(0, 1)$ distribution.

Stating that $\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}$ has an approximate $N(0, 1)$ distribution is equivalent to stating that $\bar{X}$ has an approximate $N(\mu, \sigma^2/n)$ distribution.

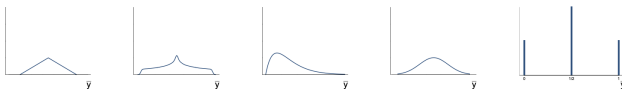
Note that the approximate normality of the sampling distribution of $\bar{X}$ applies no matter what the shape of the population distribution (the distribution that generates each individual observation) is.

Central Limit Theorem

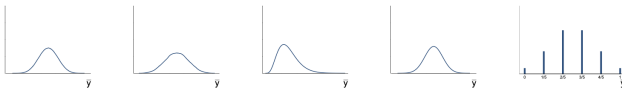
Population distributions



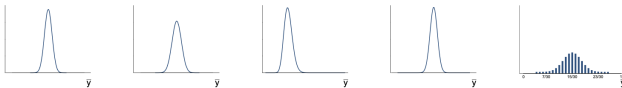
Sampling distributions of $\bar{Y}$
 $n = 2$



$n = 5$



$n = 30$



Example: Predicting Election Result from Exit Poll

Let's return to the exit poll example and apply the central limit theorem. We are interested in the sampling distribution of $\bar{X}$, the proportion of voters who vote for Candidate A.

- In the exit poll of $n = 2271$ voters, a proportion of 0.655 reported voting for Candidate A
- In the election, a proportion of 0.635 of 17 million voters reported voting for Candidate A. We can model the vote of each individual in the population as a draw from a distribution with probability 0.635 of corresponding to a vote for Candidate A and probability $1 - 0.635 = 0.365$ of corresponding to a vote for Candidate B.
- The population distribution for the 17 million voters is thus a Bernoulli distribution with success probability $p = 0.635$ (denoted by $\text{Ber}(p)$ and equivalent to a $\text{Bin}(1, p)$ distribution).
- The sampling distribution of $\bar{X}$ is approximately normal with mean $p = 0.635$ and standard deviation $\sqrt{p * (1 - p) / n} = \sqrt{(0.635 * 0.365) / 2271} \approx 0.0101$.

Example: Predicting Election Result from Exit Poll

On the other hand, if exactly half of the voters had voted for Candidate A, then:

- The population distribution would be $\text{Ber}(0.5)$
- The sampling distribution of $\bar{X}$ would be approximately normal with mean $p = 0.50$ and standard deviation $\sqrt{p * (1 - p) / n} = \sqrt{(0.635 * 0.365) / 2271} \approx 0.0105$
- With probability ≈ 0.9973 , $\bar{X}$ would take on values within three standard errors of the mean — within $3 * 0.0105$ of 0.50 , or between 0.469 and 0.531 .
- Thus observing $\bar{X} = 0.655$ would be extremely unlikely

The Delta Method

A result similar to the central limit theorem holds more generally: many sample statistics (besides the sample mean) used to estimate population parameters have approximately normal sampling distributions, for reasonably-sized samples in studies that employ randomization.

Let $T_1, \dots, T_n, \dots$ be a sequence of random variables such that $\sqrt{n}(T_n - \theta)/\tau$ converges in distribution to a $N(0, 1)$ distribution. In other words, for large n , T_n is approximately normally distributed with mean θ and standard error $\tau/\sqrt{n}$.

- The random variables T_n can be statistics – indeed the central limit theorem is the special case $T_n = \frac{1}{n} \sum_{i=1}^n X_i$
- We are interested in the distribution of $g(T_n)$ for a smooth function $g(\cdot)$ (a function is smooth if it has continuous derivatives of all orders)

The Delta Method

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- We are interested in the distribution of $g(T_n)$ for a smooth function $g(\cdot)$ (a function is smooth if it has continuous derivatives of all orders)

The **delta method** allows us to find this distribution by differentiating $g(\cdot)$:

- $\sqrt{n}(g(T_n) - g(\theta))$ converges in distribution to a $N(0, [g'(\theta)]^2 \tau^2)$ distribution

Delta Method: Example

A common use of the delta method is to find transformations that are variance stabilizing in the sense that they produce random variables with variances that are approximately the same for all values of a parameter of interest.

As an example, let X_n denote the number of telemarketing phone calls you receive in n days. We view X_n as a Poisson process with mean $\mu = n\lambda$ for some fixed $\lambda > 0$ that represents the daily rate.

- In other words, each X_n has a Poisson distribution with mean equal to $n\lambda$.

The random variable $T_n = X_n/n$ describes the average number of calls per day.

- The distribution of T_n is approximately normal with mean λ and variance λ/n .

Delta Method: Example

Consider the transformation $g(T_n) = \sqrt{T_n}$.

- Note that $g'(t) = 1/(2\sqrt{t})$
- Because the variance of T_n is approximately λ/n , the delta method implies that the variance of $\sqrt{T_n}$ is approximately

$$\frac{\lambda}{n} * \left(\frac{1}{2\sqrt{\lambda}} \right)^2 = \frac{1}{4n}.$$

- It follows that the variance of $\sqrt{n}\sqrt{T_n} = \sqrt{X_n}$ is approximately $1/4$. Thus, for a large mean, the approximate variability of $\sqrt{X_n}$ will remain constant.

Reference

Agresti, A., & Kateri, M. (2021). Foundations of Statistics for Data Scientists: With R and Python (1st ed.). Chapman and Hall/CRC.